Quiz — 1.1.3 Input, Output and Storage Devices — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (8 marks — 1 each)

1. **C** — An actuator is an output device (it acts on the physical world); the others are inputs.
2. **B** — HDDs magnetise patterns on a spinning platter read by a read/write head.
3. **C** — Optical media use a laser to detect pits and lands via reflected light.
4. **B** — Flash/SSD has no moving parts, so it is faster, silent, lower power and more durable.
5. **B** — RAM is volatile, read/write, and holds running programs and data.
6. **C** — ROM holds the bootstrap/BIOS firmware (non-volatile, normally read-only).
7. **C** — Virtual memory uses secondary storage as extra RAM when physical RAM is full.
8. **B** — Tape is serial access — cheap, high-capacity, ideal for archive/backup but poor for random access.

Section B (12 marks)

9. **[2]** — 1 mark per difference (max 2): RAM is volatile / ROM is non-volatile; RAM can be read and written / ROM is normally read-only; RAM holds running programs and data / ROM holds bootstrap/BIOS firmware.

10. **[2]** — 1 mark: non-volatile = it **retains its contents even without power** (data is not lost when the device is switched off). 1 mark: any valid example — HDD, SSD, USB flash drive, SD card, CD/DVD/Blu-ray, magnetic tape.

11. **[3]** — 1 mark recommendation: **flash / SSD (e.g. external SSD or USB flash drive)**. Up to 2 marks for justification linked to the requirements: portable (no moving parts, small/light); **fast transfer** speeds suit large video files; **durable** (shock-resistant, no moving parts to damage when carried); easily **rewritable**. (Reject answers that only list generic pros without linking to the scenario; cap at recommendation mark.)

12. **[2]** — 1 mark: when RAM is full, the OS moves less-used **pages** out to secondary storage (swap/paging file) and loads them back when needed, so secondary storage acts as extra RAM. 1 mark drawback: secondary storage is **much slower than RAM**, so heavy paging slows the system / can cause disk thrashing.

13. **[1]** — When excessive paging occurs (pages constantly swapped in and out of disk), the system spends most of its time moving pages rather than doing useful work, slowing it dramatically.

14. **[2]** — 1 mark per advantage (max 2): access from anywhere with internet; scalable capacity (pay for what you need); automatic off-site backup / disaster recovery; easy file sharing/collaboration; no need to buy/maintain local hardware.

Section C (5 marks)

15. **[5]** — Point/level marking, up to 5 for justified recommendations that **compare** the two printers against the requirements. Indicative content: - **Office → laser printer.** Fast printing of high volumes; low cost per page for text; sharp/high-quality text — matches "high volume, quick, cheap, black-and-white text". - **Home → inkjet printer.** Cheaper to buy and better for occasional colour photo printing; good colour/photo quality at low volume — matches "occasionally prints a few colour photos". - Comparison ("whereas"): a laser is more expensive to buy and less ideal for occasional colour photos, so it is overkill for the home user; an inkjet is slower and has higher per-page running costs for high volumes, so it is unsuitable for the office. - Justified link of each characteristic (speed, running cost, colour/photo quality, volume) to the stated requirement. Max 4 if the answer states pros without comparing the two printer types / does not justify against the scenario.

Total: 8 + 12 + 5 = 25